Ulysse Boucherie

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DevOps and Data Engineer modernizing Public Statistical Services

Experience

- **Ministry of the Interior** — Paris, France
  Data Scientist and IT Technical Referent — 2024 - Present

Data Engineering support, processing, securing, and automating data. Large-scale sensitive data exploitation in a secured environment (GDPR). Advanced and optimized SQL (PostgreSQL, DuckDB), Python (pandas, numpy, pyarrow, SQLAlchemy). Construction and improvement of data pipelines (ETL) with Apache Airflow. Data governance and quality control. Production of indicators for public decision-making support.

Responsible for publications on economic and financial crime

- **ETL Pipeline Redesign** — *2024-Present*
  Languages: Python, SQL, R
  Tools/Packages: Apache Airflow, Pandas, NumPy, DuckDB, PostgreSQL
  Challenges: Legacy systems with 20+ years of technical debt; Resistance to change from non-technical stakeholders; Data quality issues in source systems; Need to maintain production during migration
  Redesigned ETL processes for 20agents, reducing the delay between data acquisition and production by over 80%. Implemented incremental processing, data validation frameworks, and automated alerting. Trained team members on new processes and tools.
- **Culture of Change Implementation** — *2024-Present*
  Languages: Python, Markdown, Bash
  Tools/Packages: Git, GitHub, Docker, Jupyter, VS Code
  Challenges: Diverse audience with varying technical skills (from Excel-only to advanced Python); resistance to adopting new tools; need to demonstrate immediate value
  Implemented a culture of change with modern tools and methods. Created documentation, templates, and starter kits. Organized training sessions and workshops. Built internal tools to lower the barrier to entry for non-technical users.
- **Open-Source Publication Tools** — *2024-Present*
  Languages: Python, TypeScript, JavaScript
  Tools/Packages: React, FastAPI, Pandas, GitHub Actions, Docker
  Challenges: Balancing flexibility with standardization; ensuring reproducibility; managing dependencies; security considerations for public-facing tools
  Developed internal and open-source tools for the industrialized publication of articles and content for the general public, reducing technical debt and dependencies. Implemented CI/CD pipelines, automated testing, and documentation generation.
- **LLM for Text Clustering** — *2024-Present*
  Languages: Python
  Tools/Packages: Transformers, scikit-learn, Sentence-BERT, HuggingFace
  Challenges: Privacy constraints (NDA); handling French text with technical jargon; model selection and fine-tuning; performance optimization
  Co-designed, developed, and deployed LLM models to cluster free-text survey responses. Implemented prompt engineering, evaluated multiple model architectures, and deployed solution in production environment.
- **Economic Crime Studies** — *2024-Present*
  Languages: Python, R, SQL
  Tools/Packages: Pandas, ggplot2, dplyr, PostgreSQL, DuckDB
  Challenges: Data from multiple sources (Police Nationale, Gendarmerie) with different formats; sensitive data handling; need for reproducible analysis
  Wrote studies on economic and financial crime using data from the Police Nationale and Gendarmerie. Developed standardized analysis pipelines and visualization templates.

- **INSEE** — Nantes, France
  DevOps Engineer — 2021 – 2024

Support and exploitation of critical information systems. POC with advanced Python technologies (pandas, numpy, Dask, scikit-learn, XGBoost, TensorFlow/PyTorch, MLflow, Polars, pyarrow, Modin, Ray, FastAPI). Support and maintenance of critical information systems: complete stack in containerized environment (Docker, Kubernetes, Helm) with low-level layers (Linux Ubuntu/Debian, TCP/IP networking, S3 storage) and high-level (PostgreSQL, Redis, microservices). Analysis and resolution of internal and open-source application incidents. Continuous improvement of SI flows and processes with observability (Prometheus, Grafana, ELK, OpenTelemetry).

DevOps and support for INSEE's data science platform

- **Onyxia Platform Development** *2021-2024*
  Languages: Go, Python, TypeScript, YAML
  Tools/Packages: Kubernetes, Helm, Docker, GitLab CI/CD, GitHub Actions, ArgoCD, Vault, Prometheus, Grafana
  Challenges: Complex multi-tenant architecture; security requirements for government data; integration with legacy systems; scalability needs
  Co-developed and installed the Platform As A Service (*Onyxia*). Designed microservices architecture, implemented authentication/authorization, and deployed monitoring stack.
- **Modern Stack Implementation** *2021-2024*
  Languages: Bash, Python, YAML
  Tools/Packages: Docker, Helm, GitLab pipelines, GitHub Actions, Grafana, Prometheus, Vault, Terraform, Ubuntu/Debian, S3
  Challenges: Resistance to change from existing team; need to maintain old systems during transition; training requirements for 500+ users
  Developed and implemented a new cutting-edge technical stack. Created migration guides, automated deployment processes, and reduced deployment time from weeks to hours (10x acceleration).
- **Data Scientist Support** *2021-2024*
  Languages: Python, R, Bash
  Tools/Packages: JupyterHub, RStudio Server, PyCharm, VS Code, Kubernetes
  Challenges: Wide range of user skill levels (beginners to advanced); diverse use cases (notebooks, dashboards, IDEs); resource constraints
  Provided support to data scientists, from beginners to advanced, for integrating their services (notebooks, internal data visualization applications, IDEs: RStudio & PyCharm). Developed templates, documentation, and best practices.

- **INSEE** — Nantes, France
  Data Platform Engineer — 2020 – 2021

  System administrator for the internal data science platform
  - **Platform Components Deployment** *2020-2021*
    Languages: PowerShell, Batch, Python
    Tools/Packages: Windows Server, Active Directory, IIS, SQL Server
    Challenges: Legacy infrastructure with limited resources; security constraints; need for high availability; complex user management
    Implemented components enabling 500INSEE data scientists to save up to 80% of their time in deploying internal tools and studies. Automated deployment scripts and created self-service portals.
  - **Resource Monitoring System** *2020-2021*
    Languages: Python, PowerShell
    Tools/Packages: Prometheus, Grafana, Windows Performance Counters
    Challenges: Limited visibility into resource usage; shared physical resources; competing priorities; need for historical data
    Set up monitoring and supervision of processes to manage shared physical resources (Windows Server). Optimized resource usage by 20%. Implemented alerting and capacity planning tools.
  - **Code Translation Initiative** *2020-2021*
    Languages: Python, R
    Tools/Packages: Pandas, tidyverse, reticulate
    Challenges: 20-year old proprietary codebase; limited documentation; resistance from users comfortable with old system; compatibility requirements
    Initiated projects to translate code to move away from a proprietary data analysis license that had been in place for 20 years. Developed migration tooling and provided training on open-source alternatives.

- **Central Statistics Office** — Dublin, Ireland
  End-of-studies Internship — 2020
  Statistical Assistant
  - **Annual Statistics Report** *2020*
    Languages: R, LaTeX
    Tools/Packages: knitr, ggplot2, dplyr, R Markdown
    Challenges: Tight deadline; need for official government formatting; collaboration with multiple stakeholders; data validation requirements
    Co-edited the official report on Ireland's annual statistics. Developed automated report generation pipeline and created reusable templates for future reports.
  - **Historical Weather Data Processing** *2020*
    Languages: R, Python
    Tools/Packages: Pandas, regular expressions, OCR (Tesseract)
    Challenges: 250+ year old handwritten records; inconsistent data formats; character recognition challenges; historical date formats
    Developed specifications for the automatic reading of monthly meteorological records from over 250 years ago. Implemented prototype OCR pipeline and data validation rules.

Education

- **ENSAI** — Rennes, France
  Engineering Degree (MSc.) — 2018 – 2021
  - Computer Science specialization - Big Data - optimization - economics (micro & macro) - class delegate

- **Lycée Français Premier** Fontainebleau, France
  Preparatory Classes for Grandes Écoles 2015 – 2018
  – *MPSI – MP*: Mathematics - Physics - Engineering Sciences - Computer Science specialization - class delegate

Technical Skills

- **Programming**: Python (advanced), Java, JavaScript/TypeScript, R, Bash, Go
- **Infrastructure**: Docker, Kubernetes, Helm, CI/CD (GitLab, GitHub Actions), Terraform
- **Databases**: PostgreSQL, DuckDB, MySQL
- **Storage**: S3
- **Systems**: Linux (Ubuntu/Debian), Windows Server
- **Cloud Platforms**: Google Cloud Platform (GCP), Amazon Web Services (AWS)
- **Data Engineering**: Apache Airflow, Pandas, NumPy, Apache Spark, Dask, Polars, pyarrow, Modin, Ray
- **ML/AI**: scikit-learn, TensorFlow, Transformers, HuggingFace, LLM fine-tuning, MLflow
- **Monitoring**: Prometheus, Grafana, ELK Stack, OpenTelemetry, Datadog
- **DevOps Tools**: Ansible, Vault, ArgoCD, GitHub Workflows

Languages

- **French**: Native
- **English**: Bilingual
- **Spanish**: Fluent

Personal Projects and Interests

- **HabitatCalc** *2022-Present*
  Languages: JavaScript, TypeScript, HTML/CSS
  Tools/Packages: React, Chart.js, Bootstrap, GitHub Pages
  Challenges: Complex financial calculations; handling edge cases; responsive design for mobile; user input validation
  Real Estate Web Application: Optimization between renting and buying a property with real parameters. *HabitatCalc*.

- **TrancheMaster** *2022-Present*
  Languages: JavaScript, TypeScript, HTML/CSS
  Tools/Packages: React, D3.js, GitHub Pages
  Challenges: Complex tax bracket calculations; handling progressive tax rates; visualization challenges; edge cases in tax law
  Tax Web Application: Understanding marginal tax brackets on income and tax rates. *TrancheMaster*.

- **Conferences and Seminars**: *L'Etat dans le nuage 2026*, INSEE and Public Statistical Services Webinars 2024 to 2026, AI Salon 2023, KubeCon 2023, DevFest 2022 & 2023, VivaTech 2022
- **Sports**: Triathlon, volleyball, alpine skiing, bouldering
- **Music**: Piano (4 years), drums (10 years)